

Representing Functions and Fields

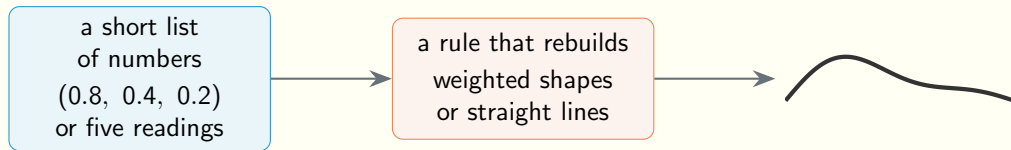
Finitely many numbers for a function, and what each choice loses

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Representing a function as a discrete list and a rule

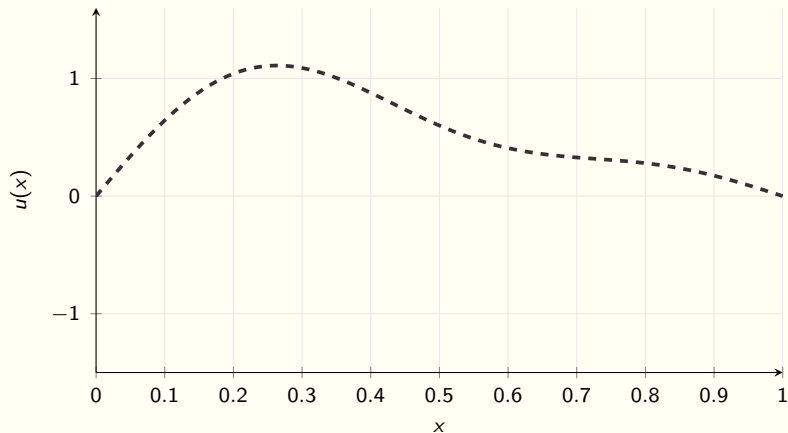


How well can finitely many numbers stand in for a function, and what decides the accuracy?

Outline

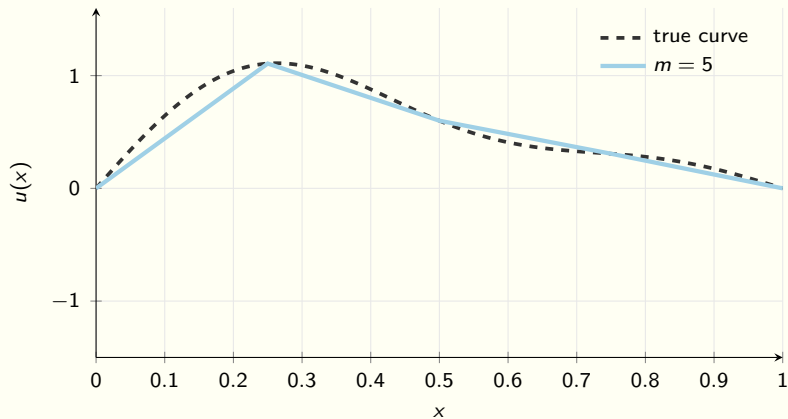
- 1 Readings at fixed stations
- 2 Coefficients on shapes
- 3 Norms and the meaning of error
- 4 Inner products, projection, and orthogonality
- 5 Changing the basis
- 6 Convergence, density, and completeness
- 7 Kernels and point evaluation
- 8 Noisy observations, ridge regression, and Gaussian conditioning
- 9 Approximating a family of functions
- 10 Adaptive representations
- 11 Choosing a representation for the cooling fin
- 12 From hats to a network

Sampling the displacement at fixed stations



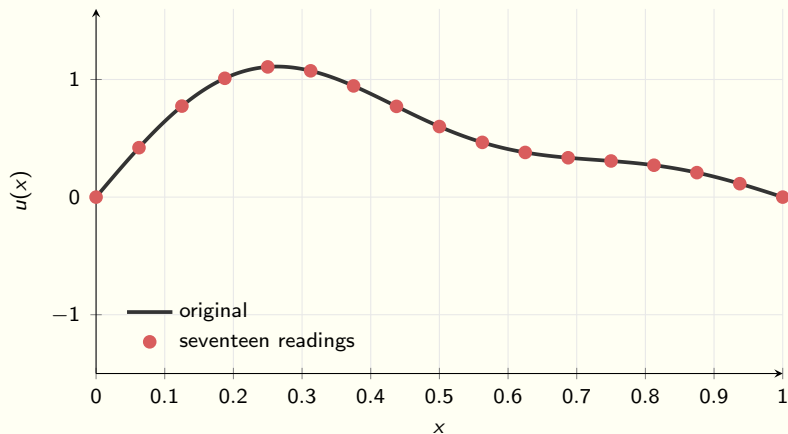
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Convergence of piecewise-linear interpolation with the spacing



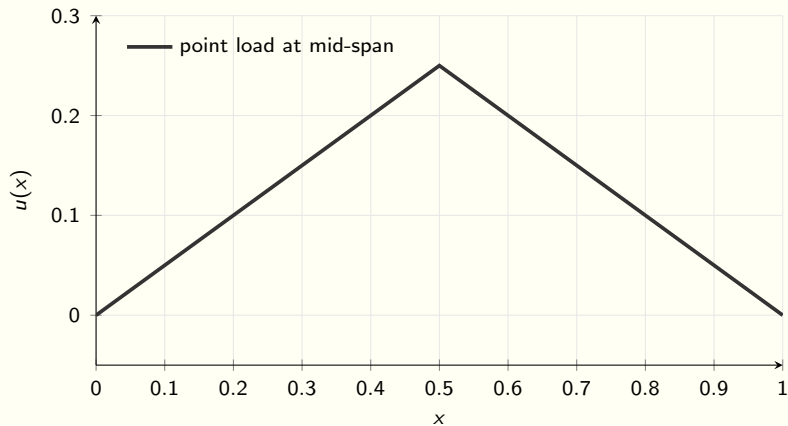
$m = 5, h = 1/4$
20.1 percent

Aliasing of a wave that vanishes at every station

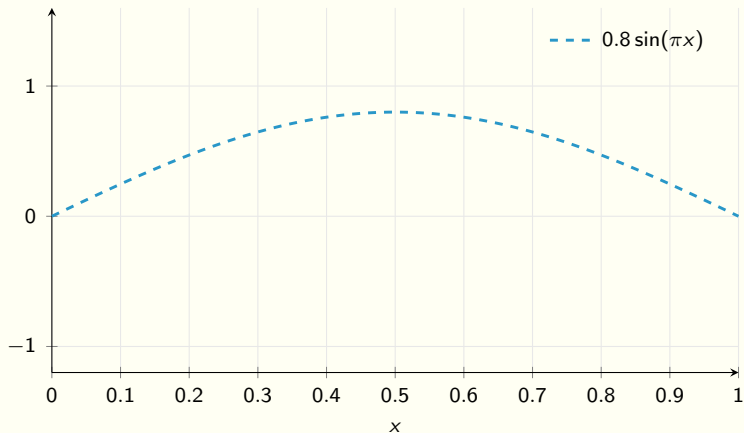


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Grid dependence of sampled representations

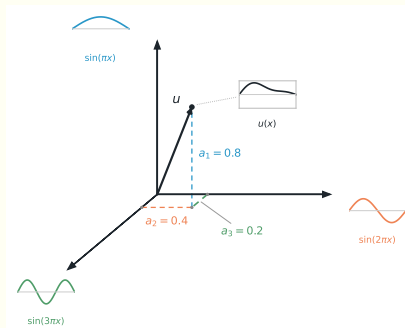
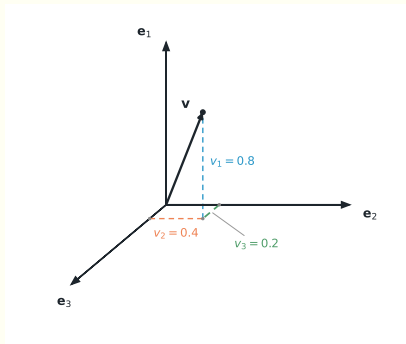


Representing a function with coefficients on sine modes



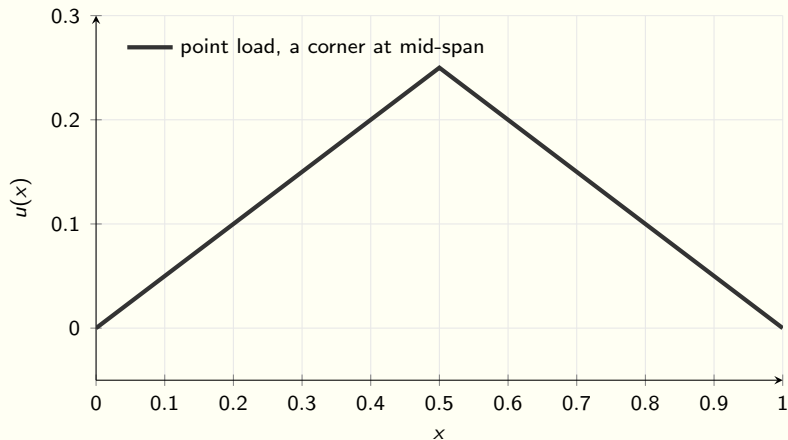
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The coefficient list as coordinates in a vector space



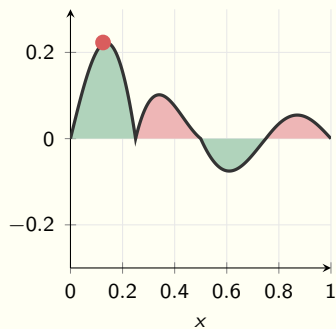
$$u + v = \sum_{i=1}^3 (a_i + b_i) \phi_i, \quad \alpha u = \sum_{i=1}^3 (\alpha a_i) \phi_i$$

Approximating a function outside the span



Measuring error in the sup, L^2 , and H^1 norms

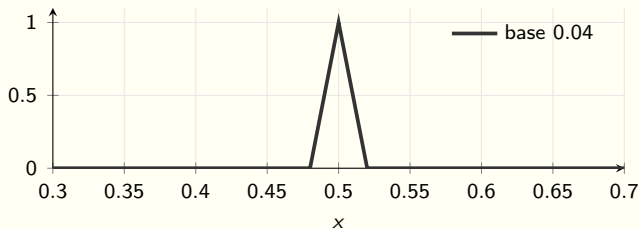
error $e = u - v$



largest gap, sup norm
0.22

[Try it live](#)

Disagreement between the sup, L^2 , and H^1 norms



spike, base 0.04

sup 1, L^2 0.115,

energy 50

[Try it live](#)

Definition of a norm and a seminorm

Norm

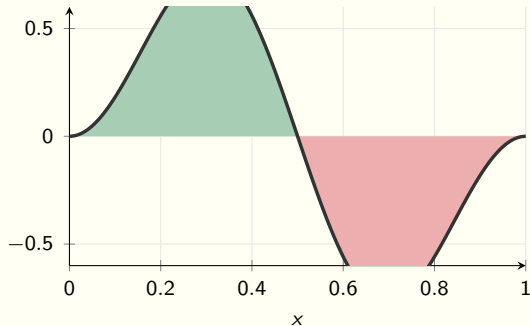
$$\|u\| = 0 \text{ only for } u = 0, \quad \|\alpha u\| = |\alpha| \|u\|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

Seminorm

$$|c|_{H^1} = \left(\int_0^1 (c')^2 dx \right)^{1/2} = 0 \quad \text{for every constant } c, \quad \text{and on the pinned string the only constant is } c = 0.$$

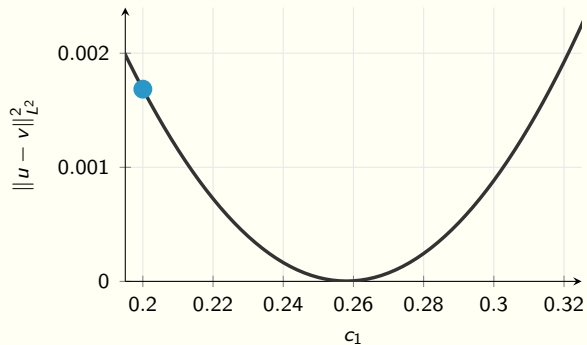
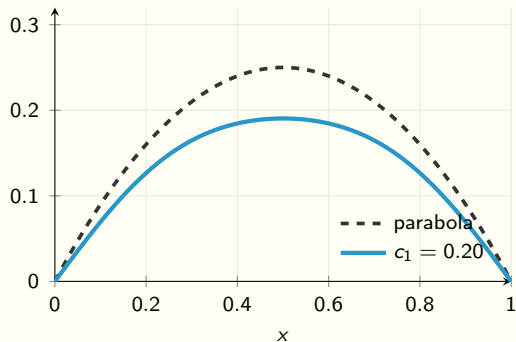
The inner product as signed overlap

$\sin(\pi x) \sin(2\pi x)$, net area 0



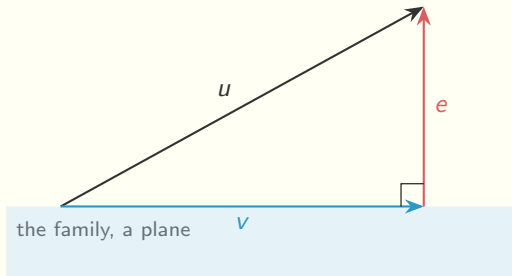
$$\langle f, g \rangle = \int_0^1 f(x) g(x) dx$$

Minimizing squared error via orthogonal projection



[Try it live](#)

Uniqueness of the projection and the Pythagorean identity



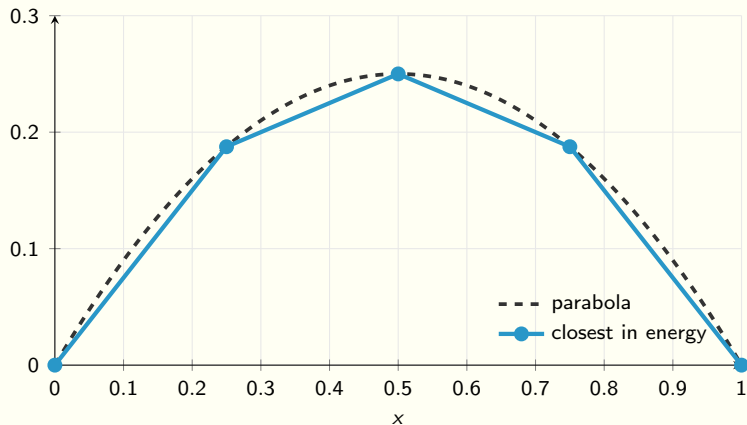
The projection theorem

Projection theorem

In a space with an inner product, a finite-dimensional family has one closest member to each target, and that member is the one whose error is orthogonal to the whole family.

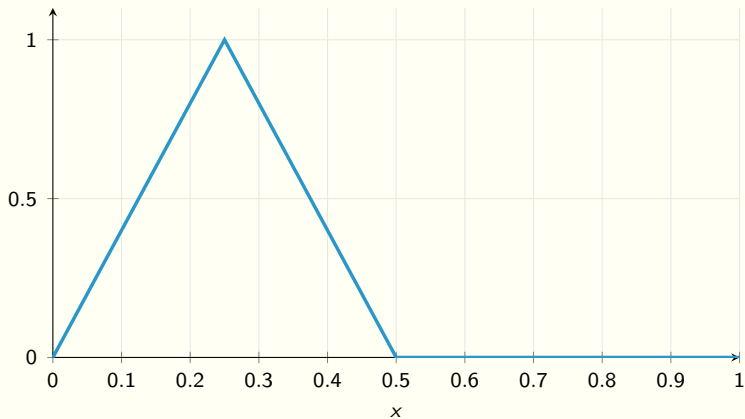
$$\langle u - v, \phi_j \rangle = 0 \quad \text{for every shape } \phi_j, \quad c_k = \frac{\langle u, \phi_k \rangle}{\langle \phi_k, \phi_k \rangle} \text{ when the shapes are orthogonal.}$$

Projection in the L^2 and energy inner products

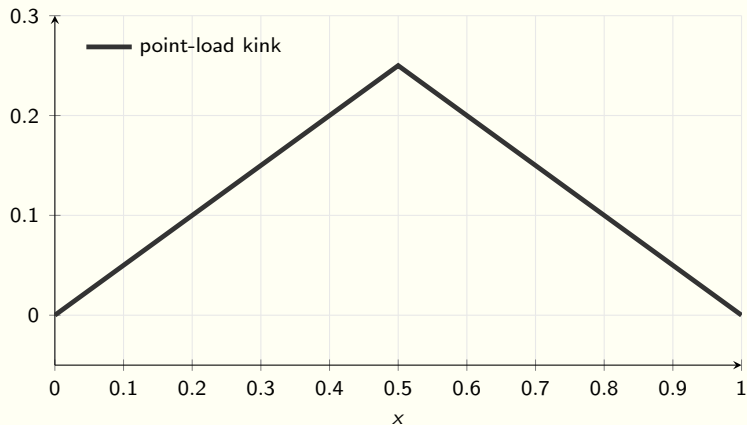


closest in energy, heights
0.1875, 0.25, 0.1875

Piecewise-linear interpolation as a hat-function expansion

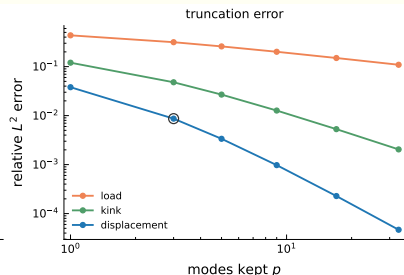
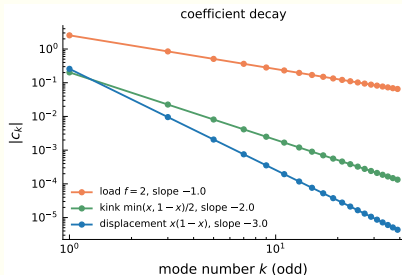


Representing a kink in the hat basis and the sine basis



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Decay rates of sine coefficients

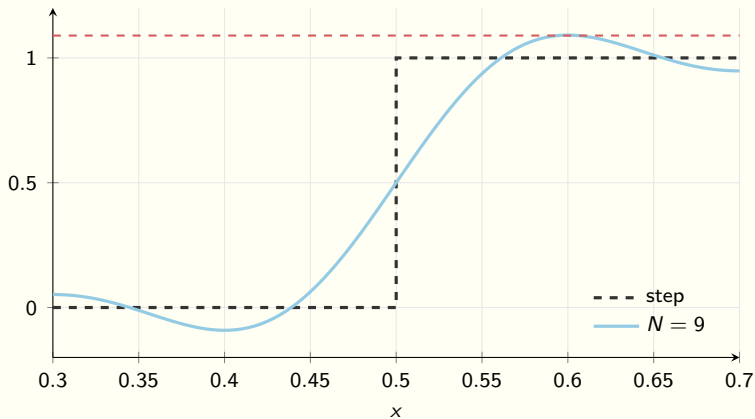


load $f=2$
 $c_k = 8/(k\pi)$, slope
 -1.0

kink
 $c_k = 2/(k\pi)^2$, slope
 -2.0

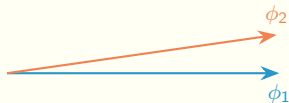
parabola
 $c_k = 8/(k\pi)^3$, slope
 -3.0

The Gibbs phenomenon at a jump



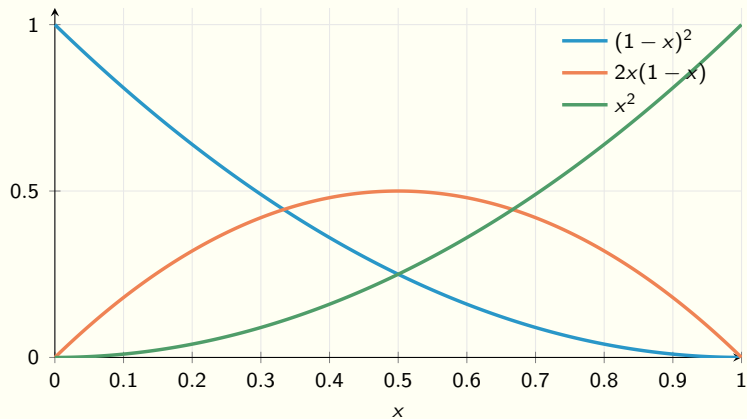
overshoot, $N = 9$
0.091

Conditioning of the Gram matrix for nearly parallel shapes



$$\mathbf{G} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \rho = 0.99, \quad \text{cond } \mathbf{G} = \frac{1 + \rho}{1 - \rho} = 199$$

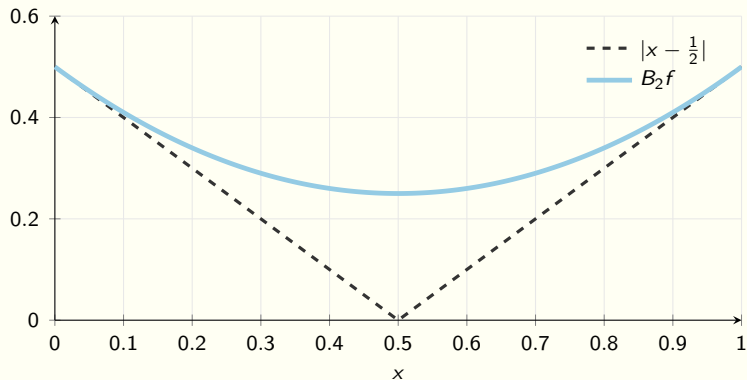
Approximating a corner with Bernstein polynomials



the three hills at mid-span

$\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

Convergence rate of Bernstein polynomials



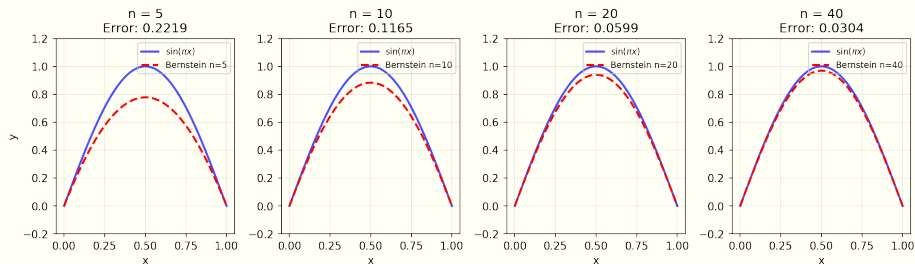
gap at the corner, $n = 2$
0.250

[Try it live](#)

$$B_n f(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$$

Bernstein basis polynomials

Bernstein Polynomial Approximation



$$B_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}, \quad B_n f(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) B_{k,n}(x)$$

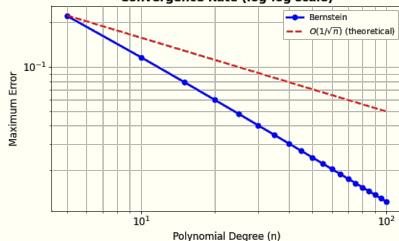
largest gap,
 $n = 5, 10, 20, 40$
0.222, 0.117, 0.060,
0.030

the basis
**nonnegative, sums
to one, peaked at
 k/n**

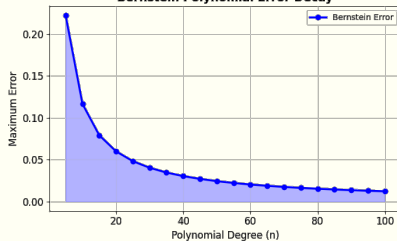
Convergence of Bernstein polynomials for a smooth function

Bernstein Polynomial Convergence: Proof of Weierstrass Theorem

Convergence Rate (log-log scale)



Bernstein Polynomial Error Decay



target
 $\sin(\pi x)$

largest gap, $n = 5$ and
 $n = 100$

0.22 and 0.013

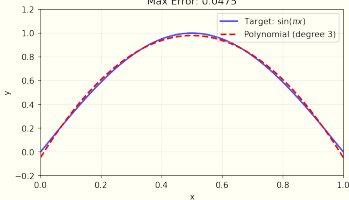
Voronovskaya at mid-span,
 $\pi^2/(8n)$

0.25 and 0.012

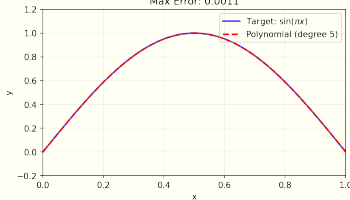
Least-squares polynomial approximation of $\sin(\pi x)$

Polynomial Approximation of $\sin(\pi x)$ (Weierstrass)

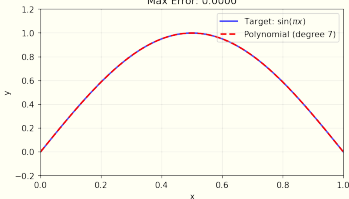
Degree 3 Approximation
Max Error: 0.0475



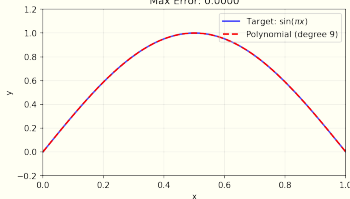
Degree 5 Approximation
Max Error: 0.0011



Degree 7 Approximation
Max Error: 0.0000



Degree 9 Approximation
Max Error: 0.0000



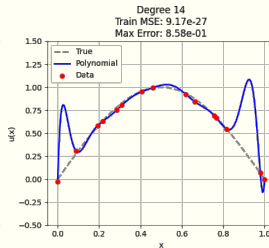
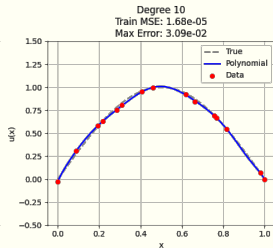
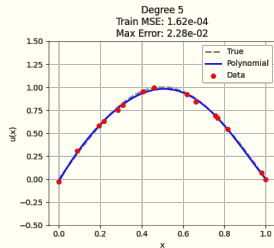
degree 3, largest gap
0.0475

degree 5, largest gap
0.0011

degrees 7 and 9, largest gap
below 0.00005

Runge's phenomenon

Runge's Phenomenon: Higher Degree $\rightarrow$ Worse Oscillations



degree 5, least squares
largest gap 0.023

degree 10, least squares
largest gap 0.031

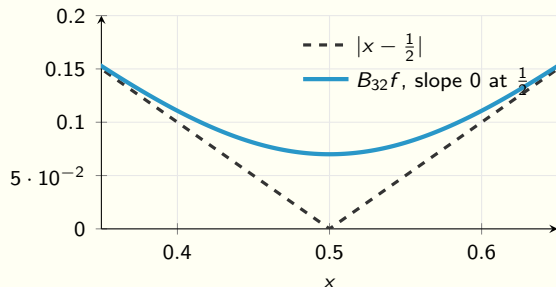
degree 14, through all
fifteen readings
**training mean
squared error**
 9×10^{-27} , **largest
gap 0.86**

The Weierstrass approximation theorem and its limits

Weierstrass (1885)

For every continuous f on $[0, 1]$ and every tolerance $\epsilon > 0$ there is a polynomial p with $|f(x) - p(x)| < \epsilon$ at every point of the interval.

$B_{32}f$ beside the corner



Bernstein's error

falls like $1/\sqrt{n}$

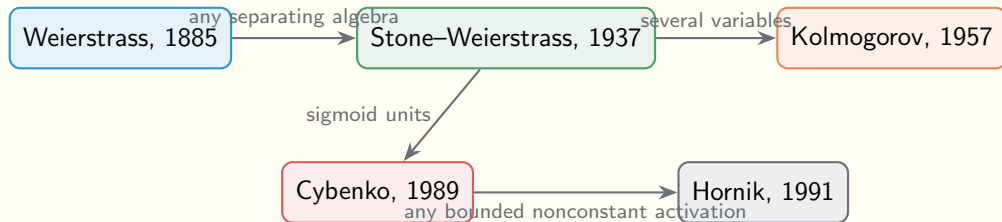
best degree- n polynomial

falls like $1/n$

the theorem names

**no polynomial, no
degree, no slope**

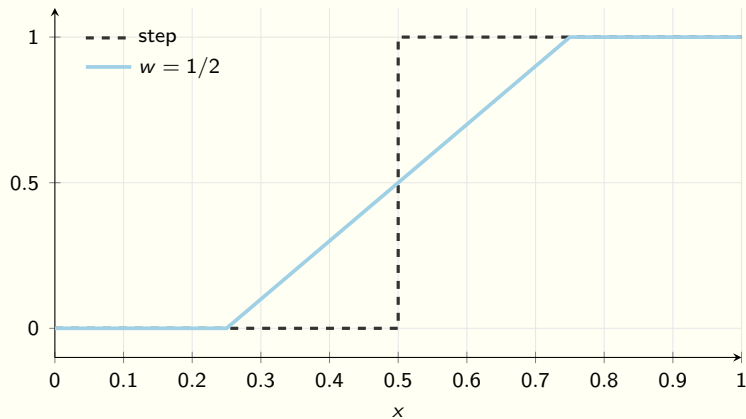
History of universal approximation theorems



what every theorem on this line gives
existence for any tolerance

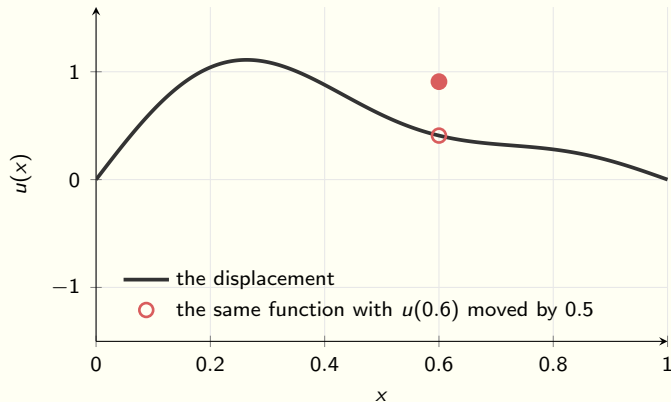
what none of them gives
a rate, a width, a slope, or a training method

Cauchy sequences, completeness, and Hilbert spaces



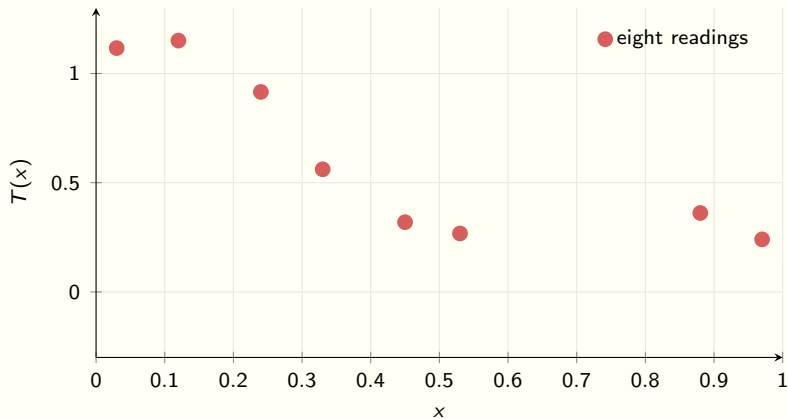
L^2 distance to the step,
 $\sqrt{w/12}$
0.204 at $w = 1/2$

Point evaluation in L^2 and H^1

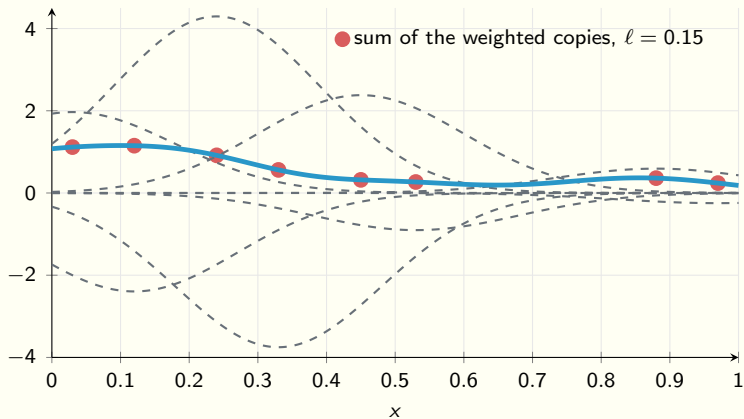


L^2 distance between the two
0

Kernels as similarity rules for scattered readings



Reconstructing a profile with weighted kernel copies



$$\mathbf{K}\alpha = \mathbf{y}, \quad K_{ij} = k(x_i, x_j)$$

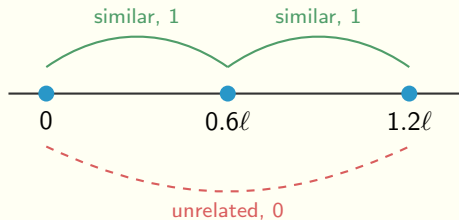
weights at $\ell = 0.15$

**alternate in sign, up
to 4.30 and -3.75**

two equal readings one ℓ
apart

0.62 each

Positive definiteness of a kernel

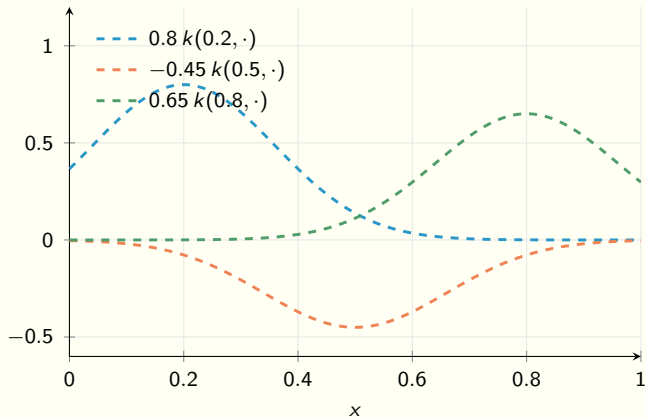


$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad \lambda = 1 - \sqrt{2}, 1, 1 + \sqrt{2}$$

$$\mathbf{w} = (1, -\sqrt{2}, 1), \quad \mathbf{w}^\top \mathbf{K} \mathbf{w} = -1.66$$

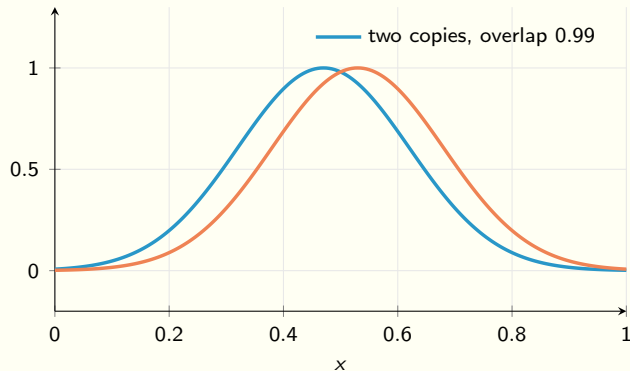
$$k(x, x') = \phi(x)^\top \phi(x') \Rightarrow \mathbf{w}^\top \mathbf{K} \mathbf{w} = \left\| \sum_i w_i \phi(x_i) \right\|^2$$

The reproducing kernel Hilbert space



$$\langle k(x_1, \cdot), k(x_2, \cdot) \rangle_k = k(x_1, x_2)$$

Noise amplification in kernel interpolation



eigenvalues of $\mathbf{K}$, overlap 0.99

1.99 along $(1, 1)$, 0.01 along $(1, -1)$

readings $(1, 1)$

weights near $\frac{1}{2}$ each

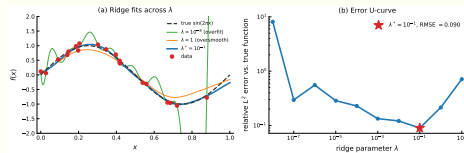
Kernel ridge regression and the choice of λ

$$\min_{f \in \mathcal{H}_k} \sum_{i=1}^n (f(x_i) - y_i)^2 + \lambda \|f\|_k^2$$

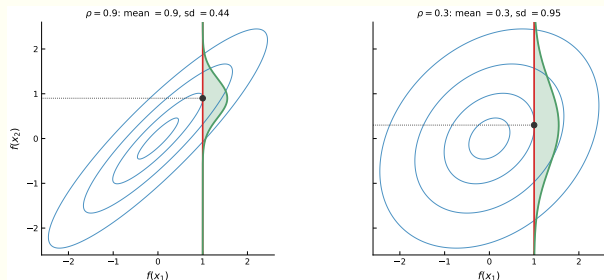
$$(\mathbf{K} + \lambda \mathbf{I})\alpha = \mathbf{y}$$

$\lambda = 0.1$, small divisor
 $0.01 \rightarrow 0.11$

amplification
a hundred $\rightarrow$ nine



Conditioning a Gaussian on an observation



$$f(x_1) = Z_1, \quad f(x_2) = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$$

$$f(x_2) \mid f(x_1) = y \sim \mathcal{N}(\rho y, 1 - \rho^2)$$

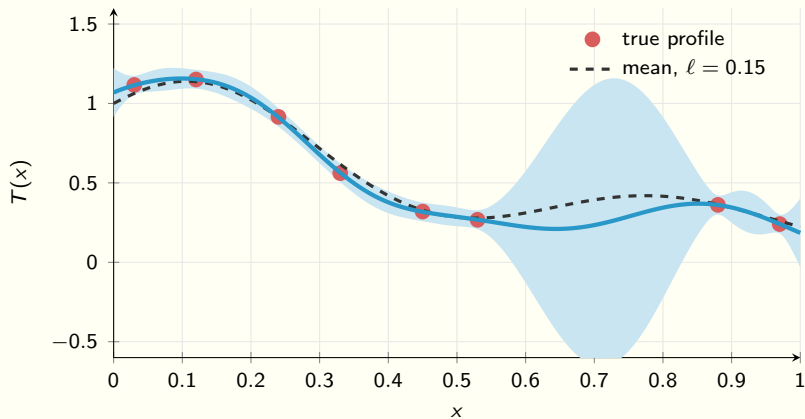
$$\rho = 0.9$$

**mean 0.9y, standard
deviation 0.44**

$$\rho = 0.3$$

**mean 0.3y, standard
deviation 0.95**

Gaussian-process reconstruction of the fin profile



mean

$$\mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$$

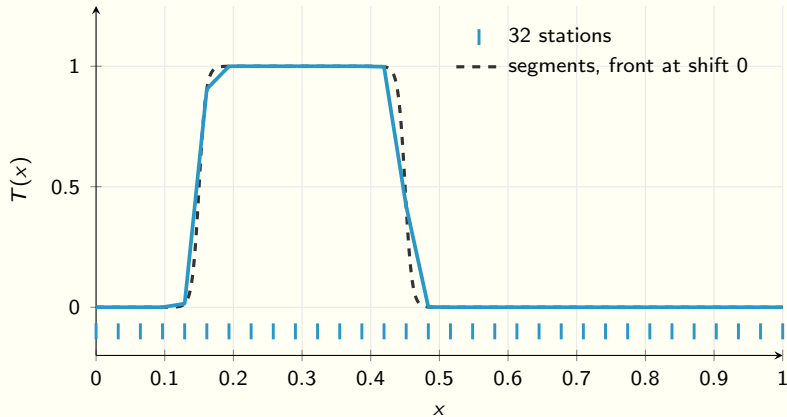
variance

$$k(x, x) - \mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(x)$$

$\ell = 0.15$

**error 10 percent,
band in the gap 0.90**

Approximating a family of functions with a fixed basis

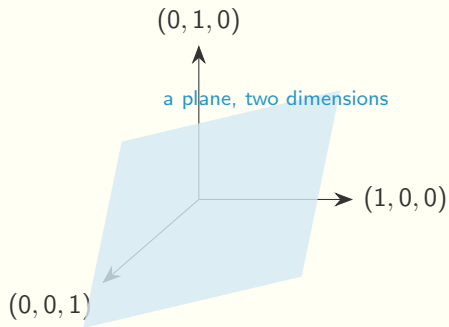


32 stations, shift 0
8.7 percent

station spacing
**three times the edge
width**

numbers that rebuild the
front at every time
**two edges, a width,
an amplitude, a shift**

The Kolmogorov n -width



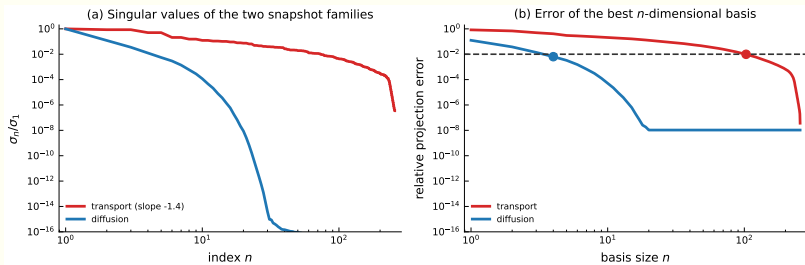
Kolmogorov n -width

$$d_n(\mathcal{M}) = \min_{\dim V=n} \max_{f \in \mathcal{M}} \min_{v \in V} \|f - v\|$$

three unit spikes, squared
distances to any plane
sum to $3 - 2 = 1$

worst spike
at least
 $\sqrt{1/3} = 0.577$

Widths of diffusion and transport families



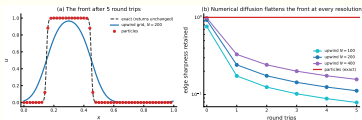
diffusing pulse, one percent
four modes

translating pulse, one percent
103 modes

coefficient k under diffusion
shrinks by $e^{-(2\pi k)^2 t}$

coefficient k under translation
changes phase only

Adaptive representations with particles, splats, and selection



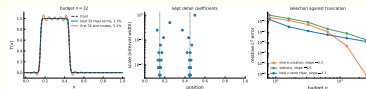
markers after five round trips
front unchanged

upwind grid, edge steepness kept
eight to sixteen percent



sixteen trained Gaussians,
96 parameters
error 0.0125

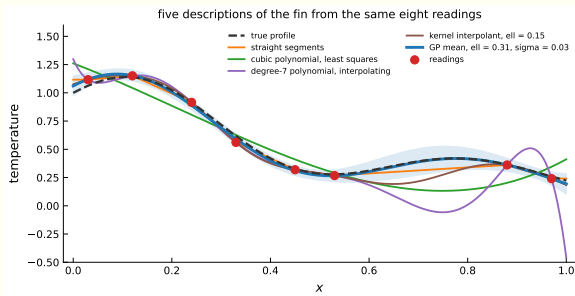
fixed grid, 100 coefficients
error 0.30



thirty-two largest Haar
coefficients
1.3 percent

first thirty-two sines
5.1 percent

Choosing a representation for the cooling fin

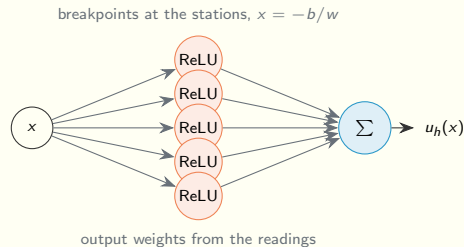
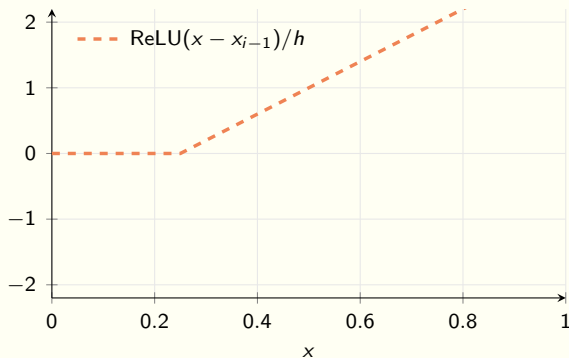


description	error vs truth	held-out
straight segments	6.5 percent	0.095
cubic, least squares	23	0.310
degree-7 interpolation	31	3.57
kernel interpolant, $\ell = 0.15$	11	0.112
Gaussian-process mean, $\ell = 0.31$	3.2	0.034
noise level		0.03

Stored numbers, shape placement, and assumption for each family

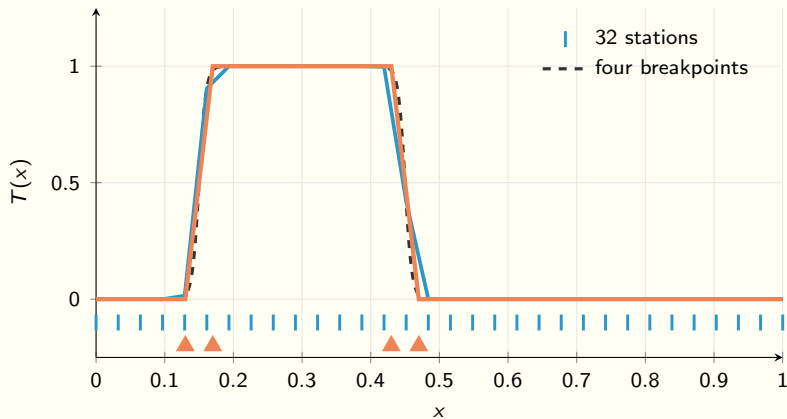
family	stores	shapes placed by	assumes
stations hats	values at fixed points nodal values	the analyst the analyst and a mesher	linear variation between stations piecewise-linear variation
sine modes kernels and Gaussian processes	global coefficients weights at the data sites	the analyst the data	a smooth odd periodic continuation smoothness at a length scale
particles Gaussian splats	positions and carried values centers, covariances, amplitudes	the dynamics the optimizer	values move with their carriers a smooth field with concentrated features
neural fields	network weights	the optimizer	the network's smoothness bias

Piecewise-linear interpolation as a one-hidden-layer ReLU network



[Try it live](#)

Movable ReLU breakpoints on a traveling front



breakpoint of
 $\text{ReLU}(wx + b)$
 $x = -b/w$

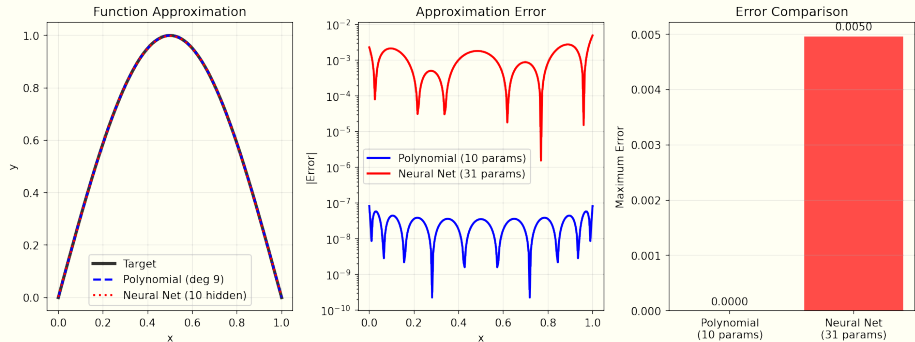
32 stations
8.7 percent

four breakpoints
4.9 percent

[Try it live](#)

Polynomial and neural network approximation of $\sin(\pi x)$

Weierstrass (Polynomial) vs Universal Approximation (Neural Network)



polynomial, ten numbers
largest gap below
 10^{-7}

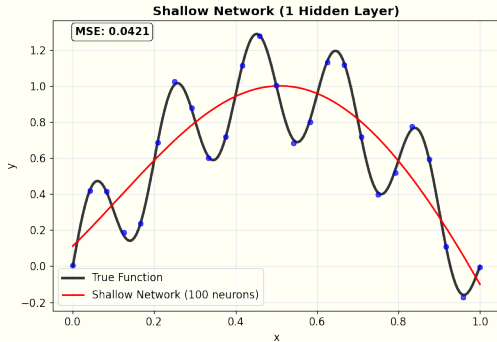
network, thirty-one
numbers
largest gap 0.0050

Polynomials and neural networks compared

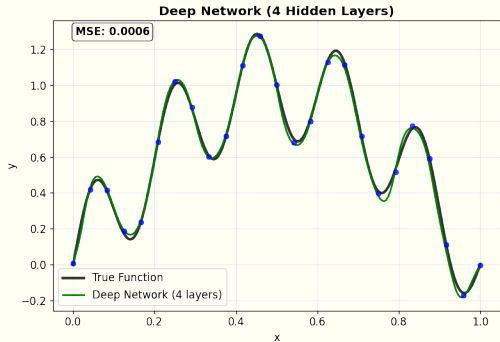
	polynomials	one-hidden-layer networks
shapes	global, every coefficient changes the curve everywhere	local ramps, each unit changes the curve beyond its breakpoint
placement	fixed by the degree	breakpoints are trained parameters
failure on sparse data	Runge oscillations near the ends	a descent that stalls
growth with dimension	coefficients grow like $\binom{n+d}{d}$ in d dimensions	parameters set by width and depth
structure	one sum	layers compose

Shallow and deep networks on a high-frequency function

High-Frequency Function: Shallow vs Deep

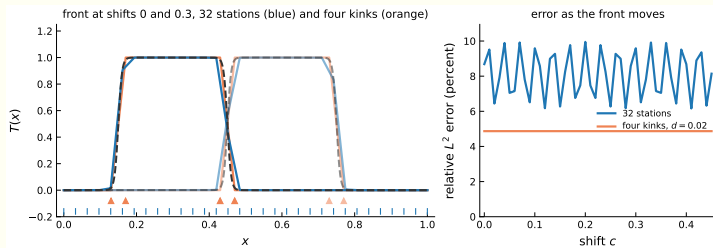


one hidden layer, a
hundred units
mean squared error
0.0421



four hidden layers
mean squared error
0.0006

From fixed shapes to trainable breakpoints



fixed breakpoints
**a linear solve, with
the projection
theorem's guarantee**

moving breakpoints
**gradient descent,
with no guarantee**

Summary

- 1 Readings at fixed stations
- 2 Coefficients on shapes
- 3 Norms and the meaning of error
- 4 Inner products, projection, and orthogonality
- 5 Changing the basis
- 6 Convergence, density, and completeness
- 7 Kernels and point evaluation
- 8 Noisy observations, ridge regression, and Gaussian conditioning
- 9 Approximating a family of functions
- 10 Adaptive representations
- 11 Choosing a representation for the cooling fin
- 12 From hats to a network